Sentinel Logic

Collusion Detection and Response: Oversight That Outlives the Transcript[1]

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Abstract

We recorded 42 file interactions between model-driven agents sharing one sandbox. A monitor reading only their declared transcript detected none of them, including the episodes in which every tool call was declared accurately and the binary answering to the declared name was hostile. Oversight of agent swarms reads a record the agent itself writes, whether that record is a transcript, a tool call, or a thinking trace. The failure is structural. Instead, we build the oversight that does not read it. A kernel probe attributes every file touch to an agent by cgroup inode, an identifier the agent cannot choose, and emits an edge when two agents touch the same file. A policy then decides whether that edge was sanctioned, so the detector records what agents did without needing to infer why. Across 80 scripted test cases spanning 19 filesystem channels, we recover 88 of 88 interactions with zero false edges, and no episode in a benign control group of 40 produces an unsanctioned edge. Agents colocated in one container are separable when the harness partitions them into cgroups. The mechanism we use covers storage channels that the kernel reports as syscalls; timing channels, network egress and asynchronous I/O fall outside coverage.

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3 Methods

4 Results

5 Discussion and Limitations

Limitations

Future Work

6 Conclusion

Code and Data

LLM Usage Statement